

Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY
Sixth Semester B.Tech Degree Examination June 2022 (2019 Scheme)

Course Code: EET302
Course Name: LINEAR CONTROL SYSTEMS

Max. Marks: 100

Duration: 3 Hours

PART A

Answer all questions, each carries 3 marks.

Marks

- | | | |
|--|--|-----|
| 1 | Explain the effect of feedback on overall gain, sensitivity and noise for a closed loop control system. | (3) |
| 2 | Explain the role of compensators in automatic control systems. | (3) |
| 3 | What is meant by bounded input bounded output stability? | (3) |
| 4 | Using Routh's stability criterion determine the stability of the given system whose characteristic equation is $s^4 + 10s^3 + 30s^2 + 100s + 25 = 0$ | (3) |
| 5 | Compare the performance characteristics of PI and PD controllers. | (3) |
| 6 | Explain the effect of adding poles and zeros to the nature of root locus. | (3) |
| 7 | Explain how phase margin and gain margin can be detected from the polar plot. | (3) |
| 8 | Draw the approximate polar plot for the open loop transfer function, | (3) |
| $G(s) = \frac{1}{s(1 + sT_1)(1 + sT_2)}$ | | |
| 9 | State and explain Nyquist stability criterion. | (3) |
| 10 | Explain the significance of Nichols chart. | (3) |

PART B

Answer one full question from each module, each carries 14 marks.

Module I

- | | | |
|----|---|-----|
| 11 | a) Derive the transfer function for the armature controlled DC motor and hence draw the block diagram representation of the system. | (7) |
| | b) Obtain the transfer function of lag compensators using R-C circuit components and bring out the characteristics of lag compensators. | (7) |

OR

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|----|--|-----|
| 12 | a) Derive the transfer function of a lag-lead compensator. | (7) |
|----|--|-----|

- b) Derive the transfer function for the field controlled DC motor and hence draw the block diagram representation of the system. (7)

Module II

- 13 a) The forward path transfer function of a unity feedback control system is given by (7)

$$G(s) = \frac{2}{s(s+3)}$$

Obtain the unit step response of the system.

- b) Derive the error coefficient and steady state error for a type 1 system applied with unit step and unit ramp input. (7)

OR

- 14 a) The open loop transfer function of a unity feedback control system is given by (8)

$$G(s) = \frac{25}{s(s+5)}$$

Determine the natural frequency of oscillation, damped frequency of oscillation, damping ratio and maximum overshoot for a unit step input.

- b) Obtain the unit step response of a standard first order system. Draw the response curve and comment on the effect of time constant on the speed of response of the system. (6)

Module III

- 15 a) A unity feedback control system has an open loop transfer function (10)

$$G(s) = \frac{K}{s(s+4)}$$

Draw the root locus and determine the value of K if the damping ratio is 0.707.

- b) Explain the Ziegler Nichols method of tuning the PID controllers. (4)

OR

- 16 a) A unity feedback control system has an open loop transfer function (10)

$$G(s) = \frac{K}{s(s+1)(s+3)}$$

Draw the root locus for the system and determine the value of K for marginal stability.

- b) Explain the function of PID controllers in any closed loop control system. (4)

Module IV

- 17 a) For the given open loop transfer function (10)

$$G(s) = \frac{10}{s(1 + 0.4s)(1 + 0.1s)}$$

Draw the bode plot using asymptotic approach; determine the phase cross over frequency, gain cross over frequency and comment on the stability of the system.

- b) Define minimum phase and non-minimum phase systems with examples. (4)

OR

- 18 a) Sketch the polar plot of a unity feedback control system having an open loop transfer function (10)

$$G(s) = \frac{1}{s(1 + s)(1 + 2s)}$$

Determine the gain margin and phase margin of the above system and comment on the stability of the system.

- b) What is the relationship between phase cross over frequency and gain cross over frequency for a stable system? How does the gain margin and phase margin values vary with these frequencies? (4)

Module V

- 19 a) Explain the general steps involved in the design of lead compensator using bode plot. (4)
- b) Design a phase lead compensator for a unity feedback system given by the open loop transfer function (10)

$$G(s) = \frac{K}{s(s + 2)}$$

to meet the following specifications

- (i) phase margin of the system $> 40^\circ$ (ii) velocity error constant is 15/sec

OR

- 20 a) The open loop transfer function of a unity feedback system is given by (10)

$$G(s)H(s) = \frac{5}{s(s + 1)(s + 2)}$$

Draw the Nyquist plot and comment about the stability of the system.

- b) What is a log magnitude Vs phase plot? (4)
